

Evaluation of Mineral Deposits

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Contents

- ▶ Mineral resource concept,
- ▶ Classification and estimation of ore reserves,
- ▶ Applications of Geostatistics and different Kriging methods

Technical aspects
Economical aspects
Social aspects

Mineral assessment is a continuous process and a number of exploration agencies like Geological Survey of India [GSI], Mineral Exploration Corporation Limited [MECL], Atomic Minerals Directorate for Exploration and Research [AMDER], State Departments of Geology and Mining [DGMs], State/Central Undertakings are engaged in exploration of mineral deposits in the country.

For mineral surveys in the country, GSI is the nodal agency to formulate exploration programmes of various agencies through Central Geological programming Board. GSI carries out mapping and systematic exploration for various mineral commodities under different stages e.g. 'reconnaissance survey' [G4], 'preliminary exploration' [G3] and 'general exploration' [G2], Detailed exploration (G1), seamlessly based on the mineral potential following the guidelines of **United Nations Framework Classification (UNFC)** and Mineral Evidence and Mineral Content Rules (MEMC-2015) with an aim to augmenting mineral resource.

Further, Indian Bureau of Mines [IBM], an attached office of Ministry of Mines collects the exploration data from various agencies. On the basis of information of exploration and the results provided by different agencies or stakeholders, IBM publishes the **National Mineral Inventory (NMI) of mineral resources in the country on an interval of once in five years**. The information of reserve/resources of minerals is being published in public domain and available on IBM website.

Under the Mines and Minerals (Development and Regulation) Amendment Act, 2015, for major minerals, the mineral concession can be allotted through auctioning. The power to grant these mineral concessions vests in the State Governments. The extraction of minerals depends on the grant of the mineral concessions by the State Government and the utilisation of minerals depends on economic viability of the minerals.

Economists, engineers, and others work together to evaluate a mineral deposit's economic potential by comparing the expected revenues from mine production with the associated expected costs of further exploration, development, and production.

It is their task to determine if the expected revenues from developing a deposit will be sufficiently in excess of costs to provide investors with an adequate return on their investment. Still other parts of the work may be *socio-economic* in nature, if a government is involved; the concern here is how the development of a deposit will affect the overall economic and social development of a region or country.

Table 6.2: Economic cut-off grades of some minerals in India and the associated conditions

S. No.	Ore/Mineral	Economic cut-off grade	Associated conditions
1.	Apatite	18% P_2O_5	Average grade should be above 20% P_2O_5 for direct use in soil reclamation.
2.	Bauxite	40% Al_2O_3	SiO_2 should be less than 5%
3.	Chromite	10% Cr_2O_3	Average grade should be about 20% Cr_2O_3 which is currently beneficiable.
4.	Copper ore	0.45% Cu	Average grade should be above 1% Cu; applicable to large resources mineable by open cast method.
		0.5% Cu	Average grade should be above-1% Cu; applicable to resources mineable by underground methods.

S. No.	Ore/Mineral	Economic cut-off grade	Associated conditions
5.	Fluorite	10% CaF_2 5% CaF_2	Mineral meant for beneficiation. Mineral processed by hand sorting.
6.	Gold	3 grams/tonne	Average grade should be 3.5-4.0 grams/tonne in case of shallow mines.
7.	Graphite	12% Fixed carbon	This is the minimum mill-feed grade for beneficiation plants in India, and the mineral is used after beneficiation.
8.	Kyanite and Sillimanite	40% Al_2O_3	Lumpy mineral; average grade should be at least 54% Al_2O_3 .
9.	Lead-zinc ore	3% Pb + Zn	Ore should be simple and not polymetallic; average grade should be above 5% Pb + Zn.
10.	Manganese ore	25% Mn	—
11.	Rock phosphate	11-16% P_2O_5	The cut-off varies depending upon the design of the beneficiation plants.
12.	Vanadium ore	0.8 to 1.0% V_2O_5 or 0.28 to 0.035% V	Based on limited mining experience. (in U.S.A. the cut-off grade considered is 0.7% V_2O_5 or 0.25% V)

Why classify reserves and resources

Government

- ▶ Manage the country's mineral resources
- ▶ Oversee financial operations

Industry

- ▶ Manage exploration and production processes

Corporate financial management

- ▶ Compare investment opportunities

Recoverable resources

The total deliverable mineral quantities from the start of production to the cease

- ▶ Discovered recoverable resources
- ▶ Potential recoverable
- ▶ Reserves (commercially recoverable)
- ▶ Undiscovered recoverable resources

Undiscovered resources

Undiscovered

- ▶ Unidentified potential (unmapped)
- ▶ No structural indication
 - ▶ Limited data
 - ▶ Poor quality data
 - ▶ No evaluation
- ▶ Structural indication

Identified potential (mapped)

- ▶ Lead
- ▶ Prospect

Minerals Include

- ▶ Metalliferous minerals - e.g. gold, silver, copper, lead, zinc, iron, aluminium, tungsten, uranium, mineral sands
- ▶ Non-metalliferous minerals - e.g. coal, phosphate, diamond, other gemstones
- ▶ Industrial and construction materials - e.g. sand, gravel, granite, limestone, kaolin

Types of Mineral Deposits

Mineral deposits can range from:

- ▶ Large scale (billions of tonnes) to small scale (hundreds of tonnes)
- ▶ Continuous to erratic distribution of mineralization
- ▶ High grade to low grade
- ▶ Very deep (>2-3km) to at surface

Types of Mineral Deposits

- ▶ The larger scale and less erratic the deposit, the wider the drill hole spacing required to confidently estimate resources and reserves, e.g.
- ▶ Large coal deposits - drill holes may be spaced up to several kilometers apart
- ▶ Small, very erratic gold deposits - drill holes may be spaced 10-15m apart

Mining Methods

- ▶ Surface mining (e.g. open cut, strip mining, quarry)
- ▶ Underground mining (e.g. block caving, open stoping, cut-and-fill, hand-held)
- ▶ In situ leaching

What is a Mineral Resource?

- ▶ A Mineral Resource is an *estimate of tonnage and grade* for a mineralized body, *based on sampling* of that body
- ▶ The estimate represents a realistic inventory that, under assumed and justifiable technical and economic conditions, might, in whole or in part, become economically extractable
- ▶ Portions of a deposit that do not have reasonable prospects for eventual economic extraction are **NOT Mineral Resources**

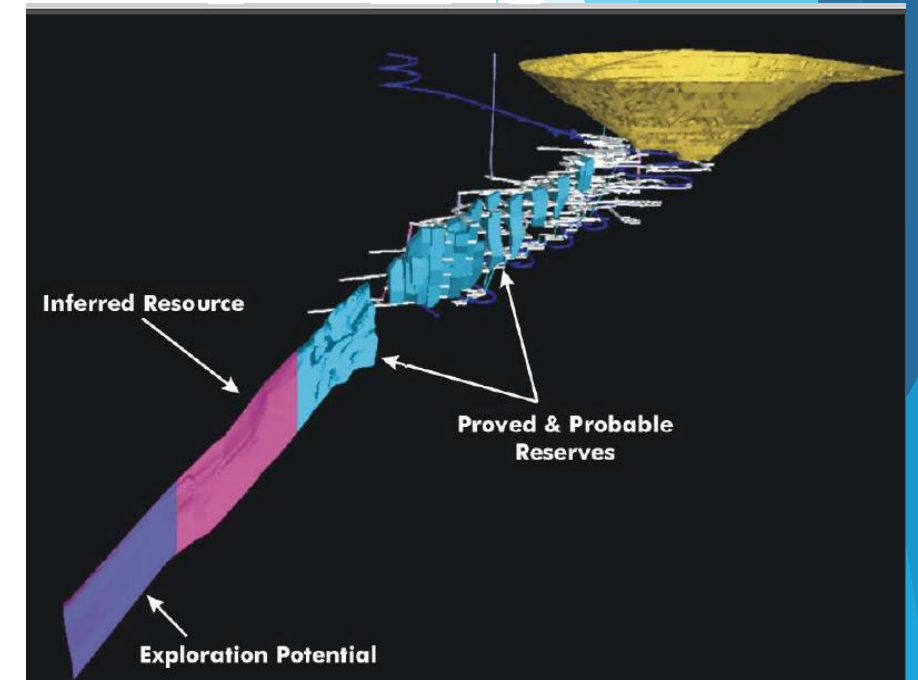
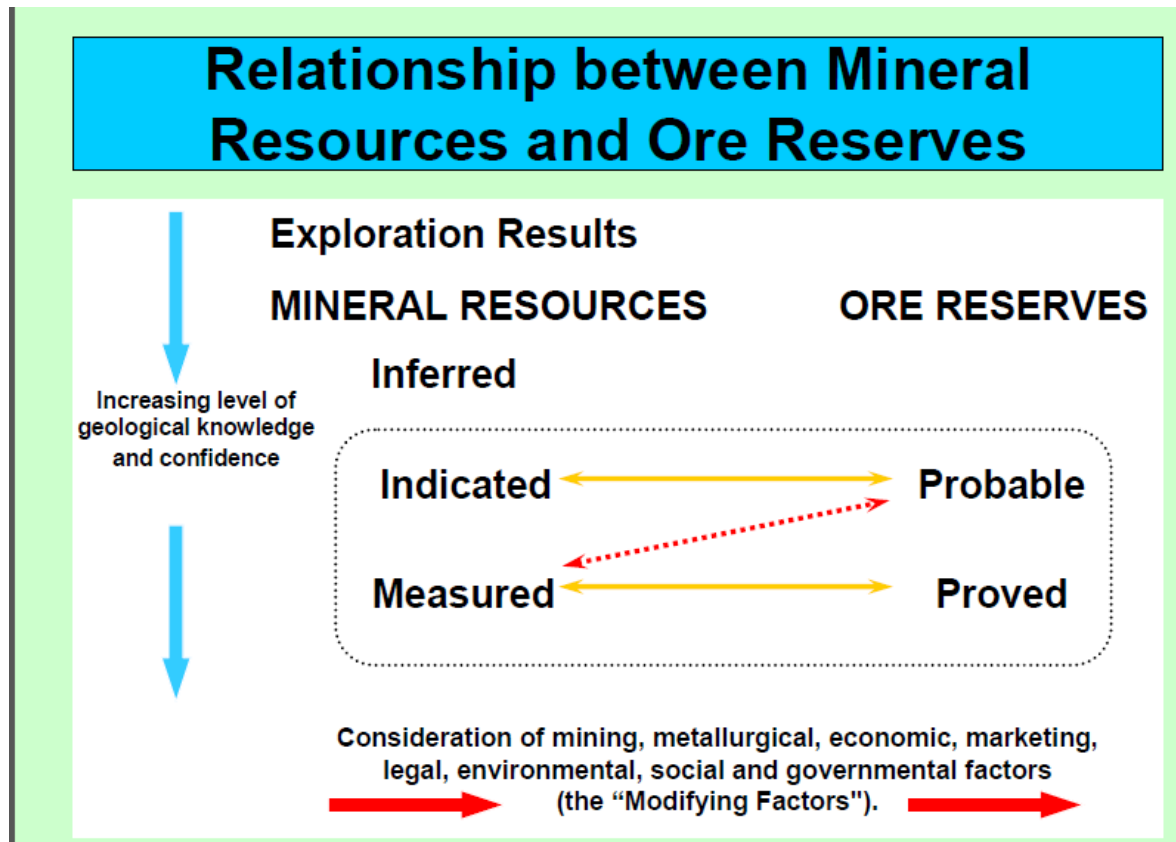
What is an Ore (Mineral) Reserve?

- ▶ An *Ore Reserve* is an *estimate of the tonnage* and grade that is expected to be delivered to the *mill or treatment* plant
- ▶ It is the economically mineable part of a Mineral Resource
- ▶ Realistically assumed Modifying Factors (*mining, metallurgical, economic, marketing, legal, environmental, social and governmental factors*) must be taken into consideration

Important Points about Mineral Resource and Ore Reserve Estimates

- ▶ Resource estimates are **ESTIMATES**, not calculations. New information or a different geological interpretation can materially change estimates
- ▶ There is no single correct resource or reserve estimate for a given deposit

Relationship between Mineral Resources and Ore Reserves



Mineral Resource Estimation

Requirements for estimating Mineral Resources:

- ▶ Confident geological interpretation
- ▶ High quality, representative samples and assays
- ▶ Application of appropriate estimation technique

This comes from:

- ▶ Mapping and sampling the deposit
- ▶ Ensuring the highest standards of sampling and assaying integrity
- ▶ Employing experienced, qualified professionals (“Competent Persons”)

Resource Estimation Techniques

These include:

- ▶ Geostatistical
 Block modelling techniques
- ▶ Statistical
- ▶ Manual;

and are applied to the geological and assaying information to yield reliable tonnage/grade estimates.

These estimates are classified and reported in accordance with one of the accepted international reporting standards (e.g. JORC, SAMREC)

Main Factors affecting Mineral Resource/ Reserve Estimates

- ▶ Reliability of geological interpretation
- ▶ Amount, distribution and quality of resource data
- ▶ In “nuggety” deposits, treatment of very high grades
- ▶ Assumptions regarding mining and treatment methods
- ▶ Assumptions regarding commodity prices and exchange rates
- ▶ Experience and judgment of Competent Person

Minerals Presentation

JORC CODE AND OTHER NATIONAL / INTERNATIONAL REPORTING STANDARDS

Characteristics of JORC-Style Reporting Standard

Principles-based standard:

- ▶ Transparency
- ▶ Materiality
- ▶ Competence
- ▶ Minimum standard for public reporting
- ▶ Classification system of tonnage/grade estimates according to:
 - ▶ Geological confidence
 - ▶ Technical/economic considerations

Characteristics of JORC-Style Reporting Standard (cont.)

- ▶ Requires public reports to be based on work undertaken by an appropriately qualified and experience person (“Competent” or “Qualified Person”), who can be held to account through a professional association
- ▶ Principle of “responsibility with accountability”
- ▶ Provides extensive guidelines on Resource/Reserve estimation

Current Mineral Resource/Reserve Reporting Standards

JORC style standards:

- ▶ JORC Code (Australasia) - translations into South American Spanish, Portuguese, Mandarin, Japanese and Russian (in progress)
- ▶ SAMREC Code (South Africa)
- ▶ CIM Standards (Canada, in NI 43-101)
- ▶ The Reporting Code (UK/Western Europe)
- ▶ SME Guidelines (USA)
- ▶ Chilean and Peruvian Codes
- ▶ CRIRSCO definitions and reporting standard template (International - in progress)

Current Mineral Resource/Reserve Reporting Standards

Other standards:

- ▶ UNECE Framework Classification (International) - incorporates JORC style definitions for market-related reporting, but is broader in scope, covering government inventory reporting
- ▶ SEC Industry Guide 7 (USA) - not fully consistent with JORC style codes

Meaning of Mineral Resource

- ▶ Concentration or occurrence of material of intrinsic economic interest in or on the Earth's crust in such form, quality and quantity that there are **reasonable prospects for eventual economic extraction**.
- ▶ Location, quantity, grade, geological characteristics and continuity are known, estimated or interpreted from specific geological evidence and knowledge.
- ▶ Sub-divided, in order of increasing geological confidence, into:
 - ▶ Inferred Mineral Resources
 - ▶ Indicated Mineral Resources
 - ▶ Measured Mineral Resources

Inferred Mineral Resource

- ▶ That part of a Mineral Resource that can only be estimated with a **low level of confidence**
- ▶ Reasons for low confidence may include:
 - ▶ Inadequate geological knowledge
 - ▶ Limited sampling data
 - ▶ Data of uncertain or poor quality
 - ▶ Uncertain geological and/or grade continuity
- ▶ “Low” in this context means usually not sufficient to allow the application of technical and economic parameters to be used for detailed planning
- ▶ Therefore Inferred Resources may not be converted directly to Ore Reserves

Indicated Mineral Resource

- ▶ That part of a Mineral Resource that can be estimated with a **reasonable level of confidence**
- ▶ “Reasonable” in this context means sufficient to allow the application of technical and economic parameters, and to enable an evaluation of economic viability
- ▶ Therefore Indicated Resources may be converted directly to Ore Reserves

Measured Mineral Resource

- ▶ That part of a Mineral Resource that can be estimated with a **high level of confidence**
- ▶ “High” in this context means sufficient to allow the application of technical and economic parameters, and to enable an evaluation of economic viability that has a greater degree of certainty than an evaluation based on an Indicated Mineral Resource
- ▶ Therefore Measured Resources may be converted directly to the highest category of Ore Reserves

Criteria for Classifying Mineral Resources as Measured, Indicated or Inferred

- ▶ Confidence in geological and grade continuity
- ▶ Quantity and distribution of sampling data
- ▶ Quality of sampling data
- ▶ Sensitivity of the Resource estimate to additional data or changes in the geological interpretation
- ▶ Judgement of the Competent Person

Economic Assumptions

- ▶ Apply to Mineral Resources and Ore Reserves, but may be a lower threshold for Mineral Resources
- ▶ Commodity prices
 - ▶ Commonly forward-looking estimates reflecting management's reasonable and supportable short and long term expectations
 - ▶ For commodities sold under existing contracts, prices may be based on contract terms
 - ▶ In the USA, SEC currently requires that commodity prices for Ore Reserves be based on the average prices prevailing in the last three years; the mining industry is currently in debate with SEC
- ▶ Exchange rates

Non-Technical/Economic Modifying Factors

- ▶ Marketing, e.g. restricted market, need to develop a market
- ▶ Legal, e.g. tenement status, critical agreements
- ▶ Environmental, e.g. environmental impact statements, national parks
- ▶ Social, e.g. impact on local community
- ▶ Governmental, e.g. national ownership requirements, royalties

Who Prepares Estimate? - the Competent/Qualified Person Concept

- ▶ JORC-style standards require publicly reported reserve and resource information to be based on work undertaken by a Competent Person
- ▶ The Competent Person is named in the public report
- ▶ It is the Competent Person's responsibility to ensure that the estimates have been performed properly
- ▶ The Competent Person may be either an employee or a consultant

Competent/Qualified Person Concept

- ▶ A Competent Person must have at least five years **relevant experience**
- ▶ A Competent Person must be a member of a professional society that:
 - ▶ requires compliance with professional and ethical standards
 - ▶ has disciplinary powers, including the power to discipline or expel a member
- ▶ Because the Competent Person's experience is in relation to the deposit style and situation under consideration, most countries do not attempt to maintain registers of Competent Persons

Recognized Overseas Professional Organizations (ROPOs)

- ▶ Established to facilitate international reciprocity of Competent Persons and thereby promote high quality reporting across national boundaries
- ▶ Australian Stock Exchange now accepts reports from overseas Competent Persons that belong to one of 17 ROPOs
- ▶ ROPO like-systems also implemented by:
 - ▶ Canadian Securities Administrators
 - ▶ Johannesburg Stock Exchange (in progress)
- ▶ ROPO's must require compliance with professional and ethical standards and have disciplinary powers, including the power to discipline or expel a member

Uses of Mineral Resource/Reserve Estimates

- ▶ Mine planning and investment/development decisions
- ▶ Property and company valuations
- ▶ Acquisitions and disposals of properties
- ▶ Debt and equity raisings
- ▶ Financial, e.g. dépréciation, impairment etc.

Current Minerals Reporting Practice

	Australasia	Canada	S Africa	UK/W Europe	Chile	Peru	USA - SME	USA - SEC	UN-ECE FC
JORC-type standard	✓	✓	✓	✓	✓	✓	✓	X	X
Recognised by Regulator	✓	✓	✓	✓	✓	✓	X		X
Competent Person	✓	✓	✓	✓	✓	✓	✓	X	X
ROPO-type system	✓	✓	✓	X	X	X	X	X	X

UN-ECE Framework Classification

I 1- 4: Remaining in place

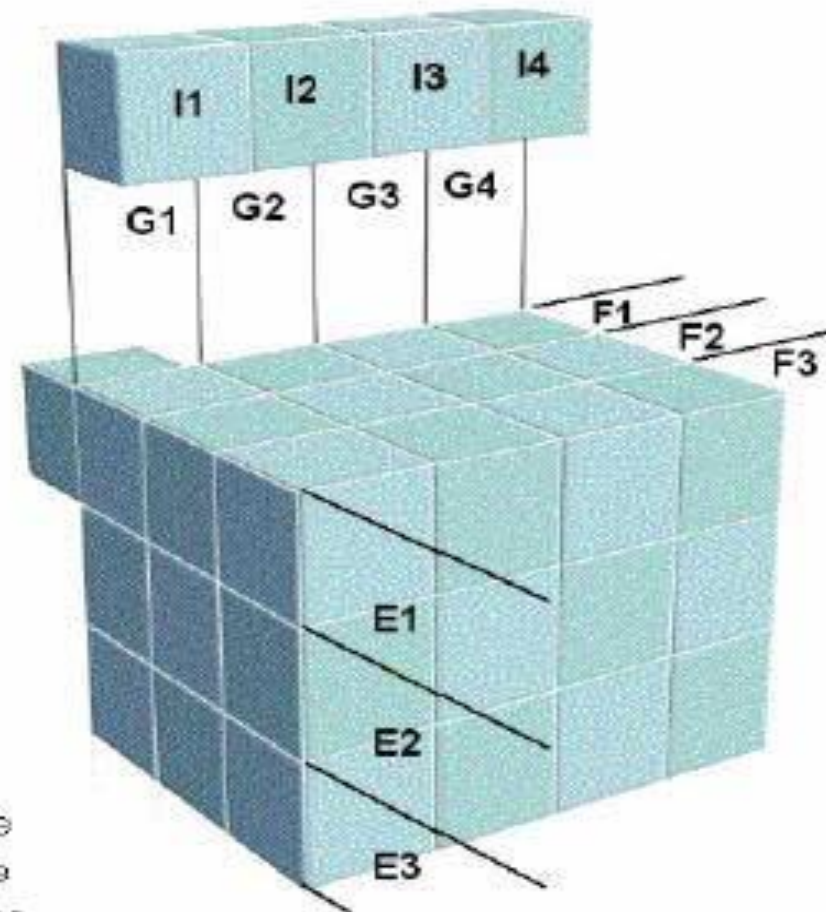
- E1:** Commercial
- E2:** Potentially commercial
- E3:** Undetermined or Non-commercial

F-1: Produced

- F1:** Committed projects
- F2:** Contingent projects
- F3:** Prospecting projects

- G1:** Reasonably assured geology
- G2:** Identified geologic conditions
- G3:** Estimated geologic conditions
- G4:** Potential geologic conditions

- I1:** Reasonably assured geology in place
- I2:** Identified geologic conditions in place
- I3:** Estimated geologic conditions in place
- I4:** Potential geologic conditions in place



Comparison of UNFC with JORC-Style Reporting Standards

UNFC	JORC-Style
3-dimensional	2-dimensional
41 possible categories*	5 possible categories
Government + market reporting	Market reporting only
No Competent Person	Competent Person
Not commonly used by “Western” banks etc	Commonly used by “Western” banks etc

*** 5 JORC-style categories incorporated into UNFC for market-related reporting**

Comparison of USA SEC with JORC-Style Reporting Standards

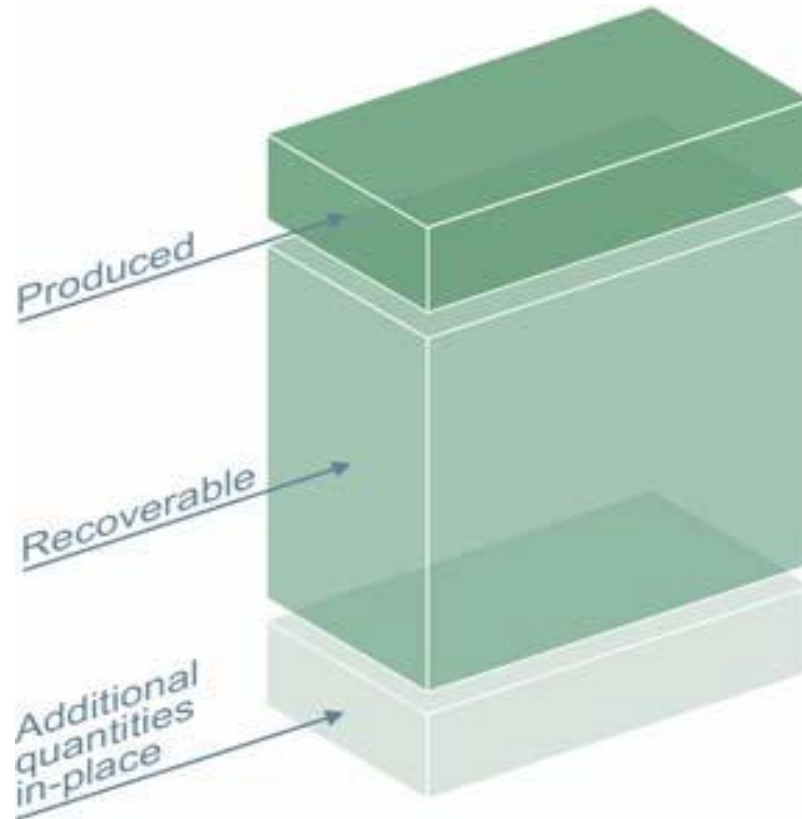
USA SEC	JORC-Style
Reserves only	Reserves and Resources
Prices 3-year average	Prices – reasonable forecasts
Feasibility study for Reserves	Technical/economic study for Reserves
No Competent Person	Competent Person

UNFC-Total Initial Resources

The total resources initially in-place of naturally occurring energy and mineral resources, are described in terms of:

- ▶ Produced quantities,
- ▶ Remaining recoverable quantities, and
- ▶ Additional quantities remaining in-place.

The main focus of the UNFC is on remaining recoverable quantities.



Total initial in-place resources

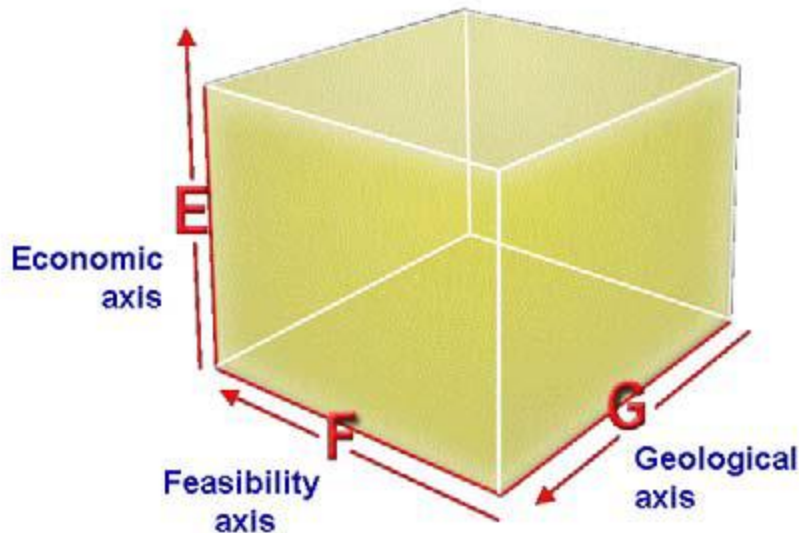
For non-renewable resources, the total resources initially in-place is constant. In inventories, material balance is therefore maintained. If any change appears, this must be explained by a re-evaluation.

Classifications

Total remaining resources are categorized using the three essential criteria affecting their recoverability:

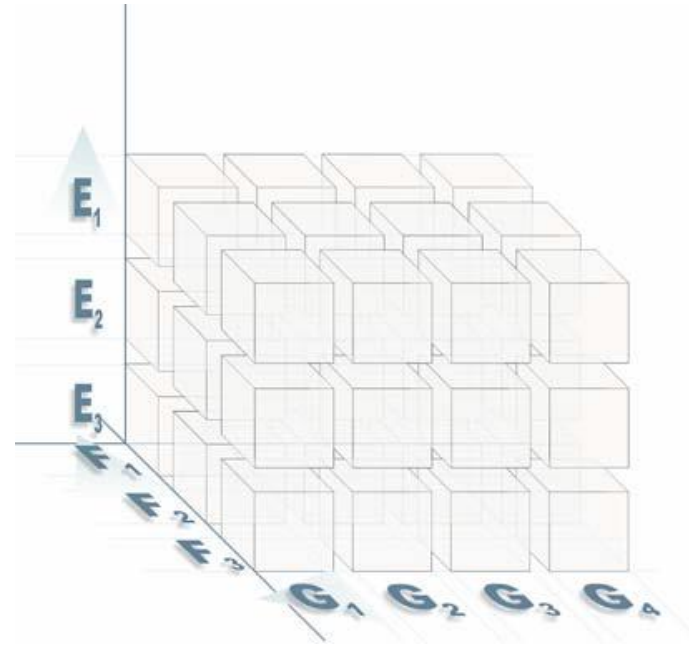
- ▶ Economic and commercial viability (E).
- ▶ Field project status and feasibility (F).
- ▶ Geological knowledge (G).

- ▶ Most of the existing resource classifications recognize these explicitly or implicitly. By making them explicit, the UNFC becomes a framework that allows for harmonization of existing classifications. The three criteria are easily visualized in three dimensions as shown in Figure



Principal elements of the UNFC

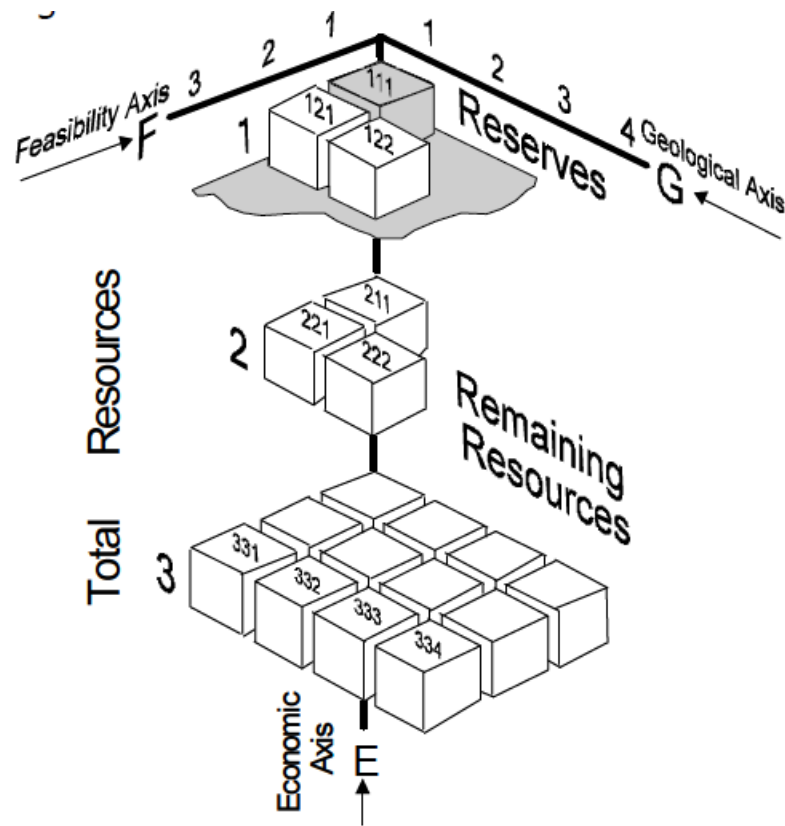
- ▶ Three main categories are used to describe economic and commercial viability,
- ▶ three to describe field project status and feasibility and
- ▶ four to describe the level of geological knowledge.



Further subdivision of the main categories is useful for special applications. Resource quantities are then grouped into classes that are defined by an E a F and a G category represented by the sub-cubes in Figure . A class of quantities may be a single sub-cube, i.e. 111, or a collection of sub-cubes. Total resources are an example of such a class where all sub-cubes are included in the class.

The three dimensions of categorization are represented by the edges of a cube. The digits are quoted in the order EFG firstly because the alphabetical order is easy to memorize, and secondly because the first digit refers to the economic viability, which is of decisive interest to producers, investors and host countries. Numbers are used to designate the different classes. Number 1, in accordance with the usual perception that the first is the best, refers to the highest degree of economic viability on the E axis, the most advanced project status on the F axis and the highest quality assessment on the G axis. The use of categories is different for fluids and for solids. This is primarily due to the fact that fluids may flow in a reservoir, irrespective of the level of geological knowledge. In the case of solids, recovery will normally be restricted to rock bodies that have been reliably assessed.

Codification



Due to variation between terminologies in different systems and languages, it is recommended to use only three-digit numeric codes for individual categories, so that they will be universally understood. For this to be possible, the sequence is always fixed, so that the quantity characterized as E1;F1;G1 may be written in number form as 111, independent of languages. In practice, only a limited number of combinations (classes) are valid. To illustrate, the UNFC for coal, uranium and other solid minerals, shown in Figure

The UNFC in matrix form applied to coal, uranium and other solid minerals

UN International Framework		Detailed Exploration	General Exploration	Prospecting	Reconnaissance
	National System				
Feasibility Study and/or Mining Report		1 (111)	usually		
		2 (211)			
Prefeasibility Study		1 (121) + (122)	not relevant		
		2 (221) + (222)			
Geological Study*)		3 (331)	3 (332)	3 (333)	3 (334)

Economic Viability Categories:

1: economic
2: potentially economic

3: intrinsically economic (economic to potentially economic)

Thanks